Notes: Kepler, McClure & Stewart (JF, 2026)
Competition Enforcement and Accounting for Intangible Capital

Contents

1 Big picture

- **Question.** Do accounting rules that exclude internally generated intangible capital from book assets weaken merger enforcement and allow anticompetitive acquisitions to avoid premerger review?
- **Core institutional feature.** Under the Hart-Scott-Rodino framework, deals in an intermediate transaction-size range face a size-of-person (SoP) test. The test depends on target and acquirer book assets and net sales. Because GAAP generally excludes internally generated intangibles, intangible-rich targets can appear small even when their economic assets are large.
- **Core empirical claim.** The exclusion of intangible capital makes thousands of deals nonreportable to the FTC and DOJ. Many of these deals are large enough to matter economically and are concentrated in technology, pharmaceuticals, chemical manufacturing, and other intangible-intensive industries.
- **Main mechanism.** The screening rule is a bright-line regulatory threshold built on accounting numbers. When book assets omit valuable customer capital, brands, patents, software, and in-process R&D, the accounting rule becomes an antitrust rule in practice.
- **Main results.** Nonreportable deals have higher deal premia, positive acquirer announcement returns when they consolidate product markets, positive rival announcement returns, and subsequent markup increases. In pharmaceuticals, nonreportable deals are more likely to involve overlapping drug projects and those projects are more likely to be discontinued.
- **Policy implication.** If intangible capital were added to the SoP test, many additional horizontal deals would become reportable. The paper estimates that the PNO would review roughly 90 additional horizontal deals per year, and that some anticompetitive deals would be deterred.
- **Takeaway.** Accounting measurement rules can have real effects outside financial reporting: they shape which mergers regulators see, and therefore which product-market consolidations can proceed without public antitrust scrutiny.

1.1 Incremental contribution of the paper

- **Beyond stealth acquisitions by deal size.** Existing antitrust-finance work shows that small deals can avoid filing thresholds. This paper shows that even economically large deals can avoid review because the regulatory asset test uses accounting book assets that omit intangible capital.
- **Accounting as enforcement infrastructure.** The paper brings financial reporting into antitrust economics by showing that GAAP-based asset measurement is not just a disclosure convention; it determines the reach of merger screening.
- **New measurement of acquired intangibles.** The authors hand-collect purchase price allocation disclosures from acquirers' 10-K filings to measure identifiable intangibles, goodwill, and tangible assets at acquisition-date fair values.
- **Product-market mechanism.** The paper does not stop at documenting nonreportability. It links nonreportable deals to deal premia, announcement returns, rivals' returns, markups, and product discontinuation, making the economic mechanism explicitly anticompetitive rather than merely administrative.
- **Policy counterfactual.** The paper estimates the enforcement implications of adding acquired intangible capital to the SoP threshold, including the number of added filings, likely horizontal reviews, potential deterrence, and budget implications.

2 Data construction and measurement

2.1 M&A sample and reportability classification

- The main M&A data cover completed U.S. acquisitions announced by publicly traded acquirers from February 2001 to February 2020.
- The authors focus on transactions subject to the SoP test: deals between the lower and upper size-of-transaction thresholds.
- A deal is classified as *reportable* if the target's book assets exceed the SoP asset threshold for that reporting year, or if the deal otherwise appears in FTC/DOJ notification evidence.
- A deal is classified as *nonreportable* if target book assets are at or below the SoP threshold in that year and the deal was not reviewed by the FTC or DOJ.
- Reportability is manually validated using public corporate filings, SEC EDGAR search, early-termination notices, and internet searches.

Interpretation: The key data construction challenge is that the legal status of a deal is not directly observed in a single commercial data set. The authors reconstruct reportability by combining deal data, threshold rules, and notification evidence.

2.2 Purchase price allocation data and intangible capital

- For completed acquisitions, acquirers allocate the purchase price across identifiable assets and liabilities after the deal.
- The authors collect values of tangible assets, identifiable intangible assets, and goodwill from purchase price allocation disclosures in 10-K filings.
- Identifiable intangibles include customer relationships and lists, patents, technology, software, trademarks, brands, in-process R&D, licenses, product rights, and related categories.
- Goodwill captures the residual value of the purchase price after identifiable net assets are recognized.
- The paper defines acquired intangible capital as identifiable intangible assets plus goodwill.

Interpretation: Purchase price allocations reveal assets that were economically valuable at acquisition but not included in the target's premerger book assets. This is exactly the accounting gap exploited by the paper's argument.

2.3 Horizontal overlap and product-market overlap

- The paper classifies horizontal deals using shared three-digit NAICS industries between acquirer and target.
- Because Refinitiv reports SIC codes, the paper maps SIC to NAICS using a crosswalk.
- ProductMarketOverlap is a more refined indicator that equals one when the acquirer and target overlap in product markets.
- The product-market overlap measure is central because the anticompetitive interpretation should be strongest when the deal consolidates overlapping markets.

Interpretation: A nonreportable deal is not necessarily anticompetitive. The product-market overlap dimension is what separates generic nonreportability from deals that plausibly reduce competition.

2.4 Outcome variables

- **Deal premia.** DealPremium is the proportion of the target's equity recognized as goodwill. It proxies for the acquirer paying for expected rents and synergies.
- **Announcement returns.** AnnReturn is the acquirer's five-day market-adjusted cumulative abnormal return centered on the announcement date. RivalReturns is the analogous return for industry rivals.
- **Markups.** Markup measures the acquirer's ability to price above marginal cost, following the intuition that markups are a direct proxy for market power.
- **Pharmaceutical project overlap.** Project overlap is based on Cortellis data on drug markets, mechanisms of action, and clinical phases.
- **Project discontinuation.** ProjectDisc'd equals one when a drug development project is terminated after acquisition.

Interpretation: The outcomes are layered. Premia and stock returns capture market expectations; markups capture realized product-market power; drug discontinuation captures a direct strategic action in undeveloped product markets.

3 Empirical specifications

3.1 Deal premium specification

The baseline deal-premium specification is:

$$DealPremium_{i,t} = \alpha + \beta Nonreportable_{i,j,t} + \tau_t + \gamma_{k(i)} + \varepsilon_{i,t}. \tag{1}$$

- $DealPremium_{i,t}$ is the proportion of target equity recognized as goodwill.
- $Nonreportable_{i,j,t}$ indicates that target assets are below the SoP threshold and the deal was not reviewed.
- τ_t are reporting-year fixed effects.
- $\gamma_{k(i)}$ are acquirer-industry fixed effects.

Interpretation: If nonreportable deals convey anticompetitive rents to acquirers, acquirers should be willing to pay more. A higher goodwill share is interpreted as consistent with expected rents from consolidation.

3.2 Announcement-return specification

For acquirer returns, the event-study regression is:

$$AnnReturn_{i,[-2,2]} = \alpha + \beta Nonreportable_{i,j,t} + \Theta Controls_{i,k,t} + \tau_t + \gamma_{k(i)} + \varepsilon_{i,t}. \tag{2}$$

- $AnnReturn$ is the five-day market-adjusted abnormal return around the announcement.
- Controls include payment method, deal premium, deal size, free cash flow, leverage, public target, relative size, Tobin's Q, and firm size.
- The main tests also interact $Nonreportable$ with $ProductMarketOverlap$.

Interpretation: Announcement returns test whether investors view nonreportable consolidating deals as value-increasing. The rival-return test is especially informative because competitors can benefit when industry competition softens.

3.3 Markup event-study specification

The markup specification is:

$$Markup_{i,t-3:t+3} = \beta_1 Nonreportable_{i,j,t} \times Post_{i,t:t+3} \times ProductMarketOverlap_{i,j,t} \quad (3)$$

$$+ \beta_2 Nonreportable_{i,j,t} \times Post_{i,t:t+3} + \beta_3 ProductMarketOverlap_{i,j,t} \times Post_{i,t:t+3} \quad (4)$$

$$+ \beta_4 Nonreportable_{i,j,t} + \beta_5 ProductMarketOverlap_{i,j,t} + \beta_6 Post_{i,t:t+3} + \tau_t + \gamma_{k(i)} + \varepsilon_{i,t-3:t+3}. \quad (5)$$

Interpretation: The triple interaction isolates whether markups increase after nonreportable product-market-overlap deals relative to other deals. This is the key real-outcome test of market-power effects.

3.4 Pharmaceutical project specifications

The pharmaceutical tests use project-level and deal-level regressions.

$$ProjectDisc'd_{i,j,p,t} = \alpha + \beta_1 Nonreportable_{i,j,p,t} + \beta_2 AcquiredProject_{i,j,p,t} + \beta_3 Nonreportable_{i,j,p,t} \times AcquiredProject_{i,j,p,t} + \beta \quad (6)$$

- $ProjectDisc'd$ indicates that a drug project is discontinued after acquisition.
- $AcquiredProject$ identifies overlapping projects acquired through M&A.
- ϕ_a are therapeutic-class-by-mechanism-of-action fixed effects.
- Controls include acquirer size, sales, leverage, EBITDA/assets, cash/assets, cash flow/assets, R&D, and Q.

Interpretation: The pharmaceutical setting is useful because product overlap can be observed at the project level. If nonreportable deals allow killer-acquisition behavior, overlapping acquired projects should be disproportionately discontinued.

4 Identification strategy

4.1 Institutional discontinuity: SoP threshold

- The central institutional source of identification is the SoP asset threshold, which determines whether otherwise similar deals are reportable.
- Figure 2 validates the rule mechanically: the probability of notification is near zero just below the asset threshold and jumps sharply just above it.
- This discontinuity confirms that the classification of deals as reportable or nonreportable is not arbitrary but directly tied to the accounting-based threshold.

Interpretation: The threshold does not randomize all deals, but it creates a sharp institutional link between accounting assets and regulatory visibility.

4.2 Comparing reportable and nonreportable deals within relevant cells

- The authors compare nonreportable deals to reportable deals of similar size and within the same reporting years.
- Fixed effects absorb common time-series enforcement changes and industry-specific baseline differences.

- Firm fixed effects in some specifications compare multiple acquisitions by the same acquirer, reducing concern that serial acquirers with different cultures drive the results.
- Product-market-overlap interactions identify whether effects are concentrated exactly where antitrust concerns should be strongest.

Interpretation: The identification is strongest when the coefficient appears only for nonreportable deals that consolidate product markets, because that pattern is hard to explain by generic differences between reportable and nonreportable transactions.

4.3 Event-study timing

- Announcement-return tests use narrow five-day event windows to reduce contamination by unrelated corporate news.
- Markup tests examine years before and after acquisition to assess whether markups rise after, not before, the nonreportable consolidating deal.
- Figure 7 provides a dynamic check: markup effects emerge after the deal and persist for at least several years.

Interpretation: Timing helps distinguish preexisting high market power from market power created by the transaction.

4.4 Pharmaceutical project design

- The pharmaceutical analysis compares overlapping projects in nonreportable and reportable acquisitions.
- It also compares acquired overlapping projects with the acquirer's internally developed projects.
- The TC-by-MOA fixed effects hold constant therapeutic category and mechanism of action, making the comparison more local within scientific product spaces.
- Economic-importance interactions ask whether discontinuation is stronger for Phase 3 projects or projects in concentrated markets.

Interpretation: This design addresses the possibility that nonreportable acquirers are simply firms that discontinue projects more generally. The key evidence is that the discontinuation effect is concentrated in acquired overlapping projects.

4.5 Threats and how the paper addresses them

- **Selection into nonreportability.** Nonreportable deals are smaller in book assets, but the paper shows they are economically large, similar to reportable deals in size ranges, and especially rich in intangibles.
- **Synergies rather than market power.** The product-market-overlap, rival-return, markup, and project-discontinuation results point toward market-power channels rather than generic operating synergies.
- **Measurement error in notification status.** Notification status is manually verified and the sharp jump in Figure 2 validates the classification.
- **Regulatory capacity.** The paper explicitly discusses PNO and agency resource constraints when evaluating policy implications.

- **General equilibrium behavior.** The authors acknowledge that changing the rule would alter managers' incentives to manipulate reporting status; they use the lease-accounting standard change to show that firms respond to threshold changes.

5 Tables and figures

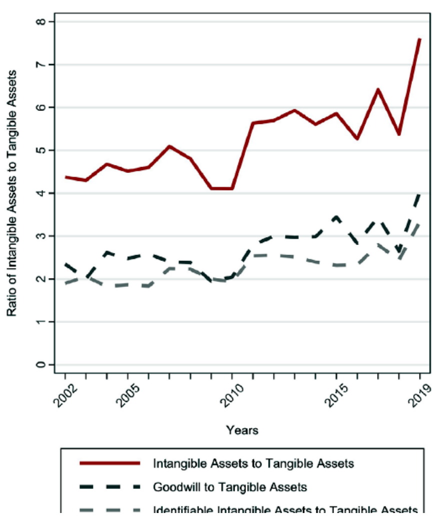
5.1 Figure 1: Ratio of acquired intangible assets to tangible assets and Figure 2: Percent of transactions that were notified

Read:

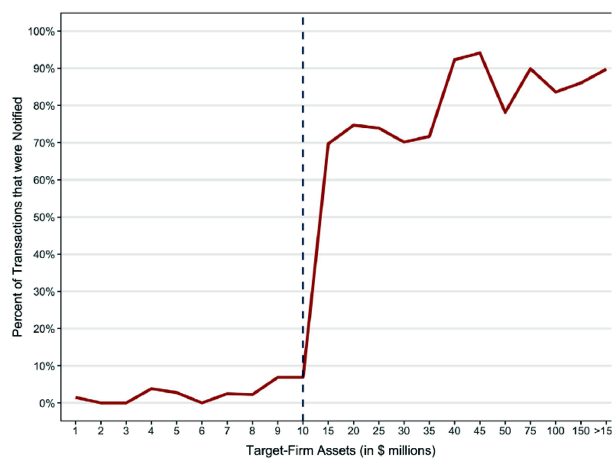
- Figure 1 shows that acquired intangibles have grown dramatically relative to tangible assets from 2002 to 2019.
- Acquired intangibles now represent roughly eight times tangible assets, with both identifiable intangibles and goodwill contributing.
- Figure 2 shows a sharp increase in premerger notification probability around the SoP asset threshold.
- The jump in Figure 2 validates that the statutory asset rule drives reportability.

Interpretation: Figure 1 establishes the macroeconomic importance of intangibles. Figure 2 shows why this importance matters for enforcement: the screening rule uses book assets, so the increasingly dominant asset class is largely omitted.

Economic message: The core problem is not only that intangibles are hard to measure. It is that an accounting omission determines which mergers antitrust agencies even see.



(a) Figure 1: Ratio of acquired intangible assets to tangible assets



(b) Figure 2: Percent of transactions that were notified

5.2 Figure 3: Notification thresholds and Table I: Descriptive Statistics

Read:

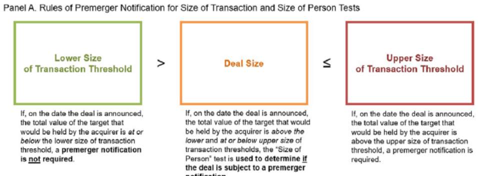
- Figure 3 explains the HSR rules: in an intermediate deal-size range, the SoP test determines whether notification is required.

- Table I shows that reportable and nonreportable deals have similar horizontal shares, but nonreportable horizontal deals are smaller in book assets.
- Nonreportable horizontal deals are concentrated in computer/electronic products and chemicals.
- Panel C shows that nonreportable horizontal deals have far more identifiable intangible assets and far fewer tangible assets than reportable deals.

Interpretation: Table I is the descriptive foundation of the paper: the transactions are not tiny or obviously irrelevant; they are economically meaningful horizontal consolidations with asset composition tilted toward intangibles.

Economic message: If antitrust screening relies on tangible-heavy accounting assets, intangible-intensive industries can consolidate more quietly.

Panel A. Rules of Premerger Notification for Size of Transaction and Size of Person Tests



Panel B. Lower and Upper and Deal-Size Thresholds (by year) and Size of Person Asset Threshold (by year)

Reporting Year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Size of Transaction Lower Threshold ($ mm)	50.0	50.0	50.0	50.0	53.1	56.7	59.8	63.1	65.2	63.4	66.0	68.2	70.9	75.9	76.3	78.2	80.8	84.4	90.0
Size of Transaction Upper Threshold ($ mm)	200.0	200.0	200.0	200.0	212.3	226.8	239.2	252.3	260.7	253.7	263.8	272.8	283.6	303.4	305.1	312.6	323.0	337.6	359.9
Size of Person Asset Threshold ($ mm)	10.0	10.0	10.0	10.0	10.7	11.3	12.0	12.6	13.0	12.7	13.2	13.6	14.2	15.2	15.3	15.6	16.2	16.9	18.0
Effective Date	Feb 1, 2001	Feb 1, 2002	Feb 1, 2003	Feb 1, 2004	Mar 2, 2005	Feb 17, 2006	Feb 21, 2007	Feb 28, 2008	Feb 12, 2009	Feb 22, 2010	Feb 24, 2011	Feb 27, 2012	Feb 11, 2013	Feb 24, 2014	Feb 20, 2015	Feb 25, 2016	Feb 27, 2017	Feb 28, 2018	Apr 3, 2019

Figure 3. Notification thresholds. The Hart-Scott-Rodino Act established the federal premerger notification program, which provides the FTC and the DOJ with information about M&A deals before they become effective. This figure depicts the premerger notification rules and threshold values used to determine whether a merger is subject to or exempt from notification. Panel

(a) Figure 3: Notification thresholds

Table I
Descriptive Statistics

This table presents descriptive statistics for our sample of reportable and nonreportable deals from February 2001 through February 2020. A deal is classified as reportable if its total assets are greater than the SoP threshold in that reporting year. A deal is classified as nonreportable if its total assets are less than or equal to the SoP asset threshold in that reporting year but has not been reviewed by the FTC or DOJ. In Panel A, we present descriptive statistics separately for reportable and nonreportable deals. In Panel B, we present descriptive statistics by industry (sorted by total deal value) for only nonreportable horizontal deals. In Panel C, we present the mean percent of tangible assets, intangible assets, and goodwill for reportable and nonreportable horizontal deals. *, **, *** represent significance at the 10%, 5%, and 1% level, respectively.

Panel A. Reportable versus Nonreportable M&As

	Reportable	Nonreportable	Difference
Type of M&A			
Horizontal (3-digit NAICS)	766 (55.2%)	219 (56.6%)	−1.0%
Nonhorizontal	621 (44.8%)	168 (43.4%)	1.0%
Average deal value (in $ millions)			
Horizontal (3-digit NAICS)	$143.5	$121.3	$22.2***
Nonhorizontal	$148.1	$122.1	$26.0***

Panel B. Nonreportable Horizontal M&As (by total deal value)

Industry	Horizontal M&As (Nonreportable)	Value (in $ billions)
Computer and Electronic Product Manufacturing	107 (48.8%)	$11.83
Chemical Manufacturing	62 (28.3%)	$8.72
Professional, Scientific, and Technical Services	17 (7.80%)	$2.11
Machinery Manufacturing	10 (4.60%)	$1.62
Telecommunications	8 (3.70%)	$0.71
Food and Kindred Products	5 (2.30%)	$0.57
Publishing Industries (except Internet)	4 (1.80%)	$0.45
Communications	5 (2.30%)	$0.44
Hospitals	1 (0.50%)	$0.12
Utilities	0 (0.00%)	$0.00
Transportation Equipment	0 (0.00%)	$0.00
Health Services	0 (0.00%)	$0.00
Merchant Wholesales, Nondurable Goods	0 (0.00%)	$0.00
Total	219 (100%)	$26.56

Panel C: Tangible Assets, Intangibles, and Goodwill of Horizontal M&As

	Reportable	Nonreportable	Difference
Horizontal M&As			
Tangible assets	35.5%	6.7%	28.8%***
Intangibles	27.7%	46.8%	−19.1%***
Goodwill	36.8%	46.4%	−9.6%***
Total	100%	100%	

(b) Table I: Descriptive Statistics

5.3 Figure 4: Trends in reported deals, Figure 5: Distribution of deals, Figure 6: Categories of intangibles, and Table II: Categories of Intangibles

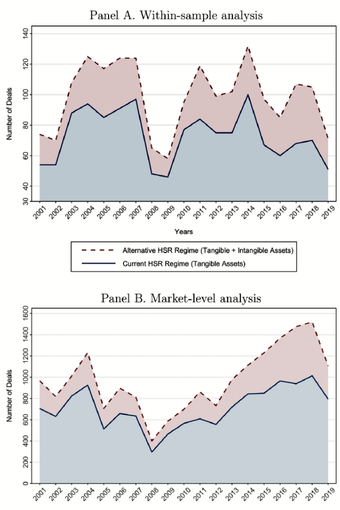
Read:

- Figure 4 shows that adding intangibles to the SoP test would substantially increase the number of reportable deals each year.

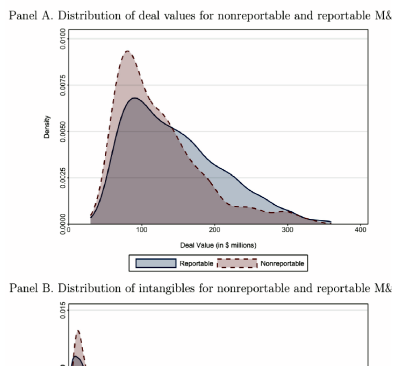
- Figure 5 compares distributions of deal values and intangible values for reportable and nonreportable deals.
- Figure 6 and Table II show that customer relationships, patents/technology/software, trademarks/brands, and in-process R&D are major categories.
- Table II shows that customer relationships account for 38.7

Interpretation: These visuals and tables convert the regulatory issue into measurable economics. The omitted assets are not vague residuals; they are concrete assets closely tied to product-market competition.

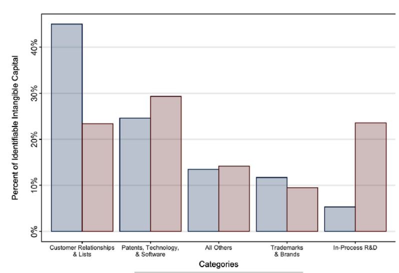
Economic message: The antitrust relevance of intangibles depends on their nature. Brands, patents, customer lists, and in-process R&D are precisely the assets that can affect pricing power, entry, and future product competition.



(a) Figure 4: Trends in reported deals



(b) Figure 5: Distribution of deals



(c) Figure 6: Categories of intangibles

5.4 Table III: Deal Premia and Nonreportable M&As

Read:

- Nonreportable deals have higher deal premia, with the baseline coefficient around 0.099.
- The interaction with ProductMarketOverlap is positive, indicating larger premia when nonreportable deals consolidate product markets.
- The result remains economically important with acquirer fixed effects, comparing deals within the same acquirer.

Interpretation: Higher premia suggest that acquirers value the private benefits of nonreportable deals. The product-overlap result strengthens the market-power interpretation because the premium is concentrated when competition is most likely to be reduced.

Economic message: Acquirers appear willing to pay more for nonreportable consolidating deals, consistent with expected rents from reduced competition.

Table II

Categories of Intangibles

This table presents results of the analysis of categories of intangibles. In Panel A, we present the frequency of intangibles in our sample. In Panel B, we present the amounts (in $ millions) and percents for all categories of identifiable intangible assets in our sample. In Panel C, we present results for difference-in-means tests by category for reportable and nonreportable deals. *, **, *** represent significance at the 10%, 5%, and 1% level, respectively.

Panel A. Frequency of Intangibles in M&A

Description	Observations
No intangibles	108
Intangibles (not disaggregated)	410
Intangibles (disaggregated by category)	1,400
Total	1,918

Panel B. Economic Importance by Category of Intangible

Category	Amount($ millions)	Percent
Customer Relationships & Lists	$30,491.91	38.7%
Patents, Technology, & Software	$19,808.12	25.1%
Trademarks & Brands	$8,906.38	11.3%
In-Process R&D	$7,663.93	9.7%
Licenses	$3,212.06	4.1%
Product Rights	$3,036.69	3.9%
Distribution Agreements	$1,242.37	1.6%
Power Purchase Agreements	$628.67	0.8%
Other Intangibles	$627.16	0.8%
Non-Compete Agreements	$513.91	0.7%
Mineral Interests	$475.20	0.6%
Usage Rights	$391.00	0.5%
Franchise Rights	$325.60	0.4%
Databases	$272.60	0.3%
Lease Intangibles	$247.96	0.3%
Supplier Agreements	$163.03	0.2%
Maintenance Contracts	$122.20	0.2%
Management Agreements	$103.10	0.1%
Pipeline Capacity Rights	$87.60	0.1%
Other Contract Rights	$66.90	0.1%
Assembled Workforce	$50.80	0.1%
Royalty Agreements	$4.90	0.0%
Total	$78,760.16	100.0%

Panel C. Difference-in-Means Tests (by Category) for Reportable versus Nonreportable M&A

Category	Mean($ millions) Reportable	Mean($ millions) Nonreportable	Difference
Customer Relationships & Lists	$25.19	$12.04	$13.15***
Patents, Technology, & Software	$13.78	$15.05	-$1.27
Trademarks & Brands	$ 6.54	$ 4.88	$1.66*
In-Process R&D	$ 2.94	$12.14	−$9.20***

Table II: Categories of Intangibles

Table III

Deal Premiums and Nonreportable M&As

This table presents results from ordinary least squares (OLS) regressions of deal premiums on an indicator for whether the deal was reportable or nonreportable to the antitrust regulators. The main variable of interest in columns (1) and (3), *Nonreportable*, is an indicator that assumes the value of one if the target firm's assets are below the size-of-person asset threshold, and zero otherwise. The main variable of interest in columns (2) and (4), *Nonreportable* × *ProductMarketOverlap*, is an interaction term that assumes the value of one when the acquirer and the target firm share overlapping product markets in a nonreportable deal, and zero otherwise. Across all columns, the dependent variable, *DealPremium*, is a continuous variable that captures the proportion of the acquired equity that is allocated to goodwill. All variables are described in Internet Appendix Section VI. We vary the inclusion of fixed effects as follows. In columns (1) and (2), we include filing-year and acquirer-industry fixed effects, respectively. In columns (3) and (4), we include filing-year and firm (i.e., acquirer) fixed effects. *DealPremium* is winsorized at the 1% and 99% levels. Per prior research (e.g., Dong et al., 2006), we exclude deals when the deal premium is larger than 200%. Robust *t*-statistics are reported in parentheses and calculated using standard errors clustered at the filing-year and the acquirer's industry level, respectively. *, **, *** represent significance at the 10%, 5%, and 1% level, respectively.

Dependent Variable:	(1) *DealPremium*	(2) *DealPremium*	(3) *DealPremium*	(4) *DealPremium*
Nonreportable	0.099***	0.084**	0.060*	0.046
	(3.16)	(2.45)	(1.86)	(1.34)
ProductMarketOverlap		−0.047*		−0.039
		(−2.15)		(−1.19)
Nonreportable × *ProductMarketOverlap*		0.049*		0.051***
		(1.91)		(4.52)
Observations	1,663	1,663	707	707
Adjusted R^2	0.151	0.155	0.481	0.482
Filing-Year F/E	Y	Y	Y	Y
Industry F/E	Y	Y	N	N
Firm F/E	N	N	Y	Y

Table III: Deal Premia and Nonreportable M&As

5.5 Table IV: Announcement Returns and Nonreportable M&As

Read:

- Panel A shows that acquirer abnormal returns are higher for nonreportable product-market-overlap deals.
- Panel B shows that rivals also earn positive abnormal returns after nonreportable deals that consolidate product markets.
- The rival result is important because rivals would not share private acquirer synergies but would benefit from softer competition.

Interpretation: Table IV provides market-based evidence. Investors price the deals as beneficial not only for acquirers but also for their rivals, which is consistent with industry-wide market-power gains.

Economic message: A procompetitive efficiency story predicts acquirer gains but not necessarily rival gains. Rival gains point toward reduced competitive pressure.

Table IV

Announcement Returns and Nonreportable M&As

This table presents results from OLS regressions of cumulative abnormal returns on an indicator for whether the deal was reportable or nonreportable to the antitrust regulators. In Panel A, the main variable of interest in columns (1) and (3), *Nonreportable*, is an indicator that assumes the value of one if the target firm's assets are below the size-of-person (SoP) asset threshold, and zero otherwise. The main variable of interest in columns (2) and (4), *Nonreportable* × *ProductMarketOverlap*, is an interaction term that assumes the value of one when the acquirer and the target firm share overlapping product markets in a nonreportable deal, and zero otherwise. Across all columns, the dependent variable, *AnnReturn*, is a continuous variable that captures the five-day market-adjusted cumulative abnormal returns of the acquirer centered on the announcement date. In Panel B, the main variable of interest in columns (1) and (3), *Nonreportable*, is an indicator that assumes the value of one if the target firm's assets are below the SoP asset threshold, and zero otherwise. The main variable of interest in columns (2) and (4), *Nonreportable* × *ProductMarketOverlap*, is an interaction term that assumes the value of one when the acquirer and the target firm share overlapping product markets in a nonreportable deal, and zero otherwise. Across all columns, the dependent variable, *RivalReturns*, is a continuous variable that captures the five-day market-adjusted cumulative abnormal return, centered on the announcement date, of the industry rivals of the acquirer. Acquirer- and deal-level control variables included, but reported in Internet Appendix Table IA.VI, in the estimations in Panel A are *AllCash*, *AllStock*, *DealPremium*, *DealSize*, *FreeCashFlow*, *Leverage*, *PublicTarget*, *RelativeSize*, *Q*, and *Size*. In Panel B, we control for *DealPremium*. All variables are described in Internet Appendix Section VI. In Panels A and B, we vary the inclusion of fixed effects as follows. In columns (1) and (2), we include filing-year and acquirer-industry fixed effects. In columns (3) and (4), we include filing-year and firm (i.e., acquirer) fixed effects. *AnnReturn* and *RivalReturns* are winsorized at the 1% and 99% levels. Robust *t*-statistics are reported in parentheses and calculated using standard errors clustered at the filing-year and acquirer-industry level, respectively. *, **, *** represent significance at the 10%, 5%, and 1% level, respectively.

Panel A. Acquirer's Announcement Returns

Dependent Variable:	(1) *AnnReturn*	(2) *AnnReturn*	(3) *AnnReturn*	(4) *AnnReturn*
Nonreportable	−0.002	−0.009	0.023**	0.008
	(−0.31)	(−1.20)	(2.95)	(0.57)
ProductMarketOverlap		0.011*		−0.015
		(2.02)		(−0.89)
Nonreportable × *ProductMarketOverlap*		0.034*		0.059**
		(2.00)		(2.32)
Observations	1,047	1,047	502	502
Adjusted R^2	0.037	0.046	0.186	0.199
Controls	Y	Y	Y	Y
Filing-Year F/E	Y	Y	Y	Y
Industry F/E	Y	Y	N	N
Firm F/E	N	N	Y	Y

(*Continued*)

(a) Panel A: Acquirer announcement returns

Table IV—*Continued*

Panel B. Rivals' Announcement Returns

Dependent Variable:	(1) *RivalReturns*	(2) *RivalReturns*	(3) *RivalReturns*	(4) *RivalReturns*
Nonreportable	0.005**	0.003*	−0.001	−0.002
	(2.70)	(2.00)	(−0.15)	(−0.73)
ProductMarketOverlap		0.001		−0.003
		(0.65)		(−0.89)
Nonreportable × *ProductMarketOverlap*		0.008*		0.007*
		(2.13)		(2.16)
DealPremium	−0.002	−0.002	0.004	0.003
	(−0.60)	(−0.62)	(0.27)	(0.25)
Observations	998	998	458	458
Adjusted R^2	0.010	0.010	0.032	0.026
Filing-Year F/E	Y	Y	Y	Y
Industry F/E	Y	Y	N	N
Firm F/E	N	N	Y	Y

(b) Panel B: Rivals' announcement returns

5.6 Table V: Markups and Intangible Capital in Nonreportable M&As and Figure 7: Markups Following Nonreportable Deals

Read:

- Table V shows that markups rise after nonreportable product-market-overlap deals.
- The dynamic terms indicate that the increase begins after the acquisition and persists for several years.

- The effect is strongest when deals involve brand- or technology-related intangibles.
- Figure 7 graphically shows the post-acquisition increase in markup coefficients.

Interpretation: Table V is the key real-outcome result. It connects nonreportable consolidation to actual pricing power rather than merely to expectations or announcement returns.

Economic message: If consumers are harmed by nonreportable consolidation, markups should rise after the deal. The table and figure provide evidence consistent with that channel.

Table V

Markups and Intangible Capital in Nonreportable M&As

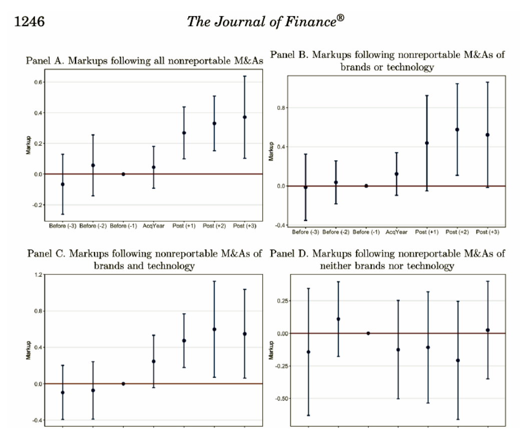
This table presents results from OLS regressions of markups on an triple-interaction term for whether the deal was reportable or nonreportable to the antitrust regulators, whether the acquirer and the target have product markets that overlap, and a time indicator. In Panel A, the main variable of interest in columns (1) and (3), *Nonreportable* × *ProductMarketOverlap* × *Post*, is an indicator that assumes the value of one if the target firm's assets are below the size-of-person (SoP) asset threshold, the acquirer and the target have product markets that overlap, and the year the markup is measured is the acquisition year (*AcqYear*) or after, and zero otherwise. The main variables of interest in columns (2) and (4), *Nonreportable* × *ProductMarketOverlap* × *Post* (+3), *Nonreportable* × *ProductMarketOverlap* × *Post* (+2), *Nonreportable* × *ProductMarketOverlap* × *Post* (+1), *Nonreportable* × *ProductMarketOverlap* × *AcqYear*, *Nonreportable* × *ProductMarketOverlap* × *Before* (−2), and *Nonreportable* × *ProductMarketOverlap* × *Before* (−3), are triple-interaction terms that include a time indicator that takes the value of one if the markup is measured three years after, two years after, the year of, one year after, two years before, or three years before the acquisition, respectively, and zero otherwise. The exclusion year is the year immediately before the acquisition year. In Panel B, the main variable of interest in columns (1), (3), and (5), *Nonreportable* × *ProductMarketOverlap* × *Post*, is an indicator that assumes the value of one if the target firm's assets are below the SoP asset threshold, the acquirer and the target have product markets that overlap, and the year the markup is measured is after the acqusiition year, and zero otherwise. The main variables of interest in columns (2), (4), and (6), *Nonreportable* × *ProductMarketOverlap* × *Post* (+3), *Nonreportable* × *ProductMarketOverlap* × *Post* (+2), *Nonreportable* × *ProductMarketOverlap* × *Post* (+1), *Nonreportable* × *ProductMarketOverlap* × *AcqYear*, *Nonreportable* × *ProductMarketOverlap* × *Before* (−2), and *Nonreportable* × *ProductMarketOverlap* × *Before* (−3), are triple-interaction terms that include a time indicator that takes the value of one if the markup is measured three years after, two years after, one year after, the year of, two years before, or three years before the acquisition, respectively, and zero otherwise. Across all columns of Panels A and B, the dependent variable, *Markup*, is a continuous variable that captures the acquirer's markup. All variables are described in Internet Appendix Section VI. In Panels A and B, across all columns, we include acquisition-year and firm (i.e., acquirer) fixed effects. *Markup* is winsorized at the 1% and 99% levels. Robust *t*-statistics are reported in parentheses and calculated using standard errors clustered at the acquisition-year and the acquirer's industry level, respectively. *, **, *** represent significance at the 10%, 5%, and 1% level, respectively. Coefficients not displayed in columns (2) and (4) of Panel A, and columns (2), (4), and (6) of Panel B, are reported in Internet Appendix Table IA.VIII.

Panel A. Markups Following Nonreportable M&As

	(1)	(2)	(3)	(4)
Dependent Variable:	*Markup*	*Markup*	*Markup*	*Markup*
Sample:	*Full Sample*	*Full Sample*	*Never- or Last-Treated*	*Never- or Last-Treated*
Nonreportable × *ProductMarketOverlap* × *Post*	0.244***		0.257***	
	(5.48)		(5.59)	
Nonreportable × *ProductMarketOverlap* × *Post* (+3)		0.352**		0.371***
		(2.75)		(2.90)
Nonreportable × *ProductMarketOverlap* × *Post* (+2)		0.321***		0.331***
		(3.96)		(3.90)
Nonreportable × *ProductMarketOverlap* × *Post* (+1)		0.264***		0.269***
		(3.34)		(3.32)
Nonreportable × *ProductMarketOverlap* × *AcqYear*		0.041		0.045
		(0.67)		(0.68)

(*Continued*)

(a) Table V Panel A: Markups following nonreportable M&As



(b) Figure 7: Dynamic markup effects

Table V—*Continued*

Panel A. Markups Following Nonreportable M&As

	(1)	(2)	(3)	(4)
Dependent Variable:	*Markup*	*Markup*	*Markup*	*Markup*
Sample:	*Full Sample*	*Full Sample*	*Never- or Last-Treated*	*Never- or Last-Treated*
Nonreportable × *ProductMarketOverlap* × *Before* (−2)		0.060		0.057
		(0.64)		(0.61)
Nonreportable × *ProductMarketOverlap* × *Before* (−3)		−0.058		−0.066
		(−0.64)		(−0.71)
Nonreportable × *ProductMarketOverlap*	−0.162	−0.163	−0.038	−0.035
	(−1.24)	(−1.08)	(−0.31)	(−0.26)
Nonreportable × *Post*	0.025		0.021	
	(0.43)		(0.37)	
ProductMarketOverlap × *Post*	−0.033		−0.046	
	(−0.58)		(−0.82)	
Nonreportable	0.003	0.023	−0.015	0.005
	(0.02)	(0.13)	(−0.11)	(0.03)
ProductMarketOverlap	−0.017	0.010	−0.054**	−0.031
	(−0.94)	(0.45)	(−2.49)	(−1.36)
Post	−0.012		−0.008	
	(−0.26)		(−0.17)	
Sample Markup (Mean)	2.284	2.284	2.278	2.278
Observations	6,748	6,748	6,643	6,643
Adjusted R^2	0.886	0.886	0.885	0.884
Acquisition-Year F/E	Y	Y	Y	Y
Firm F/E	Y	Y	Y	Y

Panel B. Markups and Intangible Capital

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable:	*Markup*	*Markup*	*Markup*	*Markup*	*Markup*	*Markup*
Subsample:	*Brand or Tech=1*	*Brand or Tech=1*	*Brand & Tech=1*	*Brand & Tech=1*	*Brand & Tech=0*	*Brand & Tech=0*
Nonreportable × *ProductMarketOverlap* × *Post*	0.407**		0.522***		−0.093	
	(2.75)		(4.19)		(−0.45)	
Nonreportable × *ProductMarketOverlap* × *Post (+3)*		0.522*		0.548**		0.025
		(2.03)		(2.35)		(0.14)
Nonreportable × *ProductMarketOverlap* × *Post (+2)*		0.576**		0.598**		−0.208
		(2.58)		(2.38)		(−0.96)
Nonreportable× *ProductMarketOverlap* × *Post (+1)*		0.438*		0.473***		−0.108
		(1.88)		(3.37)		(−0.53)
Nonreportable × *ProductMarketOverlap* × *AcqYear*		0.122		0.245*		−0.126
		(1.18)		(1.78)		(−0.70)
Nonreportable × *ProductMarketOverlap* × *Before (-2)*		0.037		−0.073		0.110
		(0.35)		(−0.49)		(0.80)
Nonreportable × *ProductMarketOverlap* × *Before (−3)*		−0.015		−0.100		−0.143
		(−0.09)		(−0.67)		(−0.61)

(*Continued*)

(a) Table V Panel B: Markups and intangible capital

Table V—*Continued*

Panel B. Markups and Intangible Capital

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable:	*Markup*	*Markup*	*Markup*	*Markup*	*Markup*	*Markup*
Subsample:	*Brand or Tech=1*	*Brand or Tech=1*	*Brand & Tech=1*	*Brand & Tech=1*	*Brand & Tech=0*	*Brand & Tech=0*
Nonreportable × *ProductMarketOverlap*	−0.164	−0.171	0.172	0.228*	0.488	0.499
	(−1.20)	(−1.12)	(1.19)	(1.79)	(0.95)	(0.94)
Nonreportable × *Post*	−0.049		−0.070		0.193	
	(−0.64)		(−0.52)		(1.03)	
ProductMarketOverlap × *Post*	−0.013		−0.151**		−0.102	
	(−0.20)		(−2.10)		(−1.02)	
Nonreportable	−0.032	−0.016	−0.115	−0.165*	−0.383	−0.360
	(−0.23)	(−0.08)	(−1.27)	(−1.94)	(−1.08)	(−1.14)
ProductMarketOverlap	0.020	0.060	0.049	0.068	−0.023	−0.024
	(0.55)	(1.04)	(0.79)	(1.38)	(−0.22)	(−0.25)
Post	−0.042		0.029		0.050	
	(−0.80)		(0.35)		(0.73)	
Observations	4,207	4,207	2,100	2,100	2,436	2,436
Adjusted R^2	0.900	0.900	0.900	0.898	0.871	0.870
Never- or Last-Treated Sample	Y	Y	Y	Y	Y	Y
Acquisition-Year F/E	Y	Y	Y	Y	Y	Y
Firm F/E	Y	Y	Y	Y	Y	Y

(b) Table V Panel B continued

5.7 Table VI: Overlapping Pharmaceutical Projects and Nonreportable M&As

Read:

- Panel A shows that nonreportable pharmaceutical deals are more likely to involve overlapping projects.
- Panel B shows that projects in nonreportable deals are more likely to be discontinued after acquisition.
- The effect is stronger for economically important projects such as Phase 3 projects or projects in more concentrated markets.
- Panel C compares acquired overlapping projects with the acquirer's own internal projects and finds elevated discontinuation for acquired overlapping projects in nonreportable deals.

Interpretation: Table VI moves from developed product markets to undeveloped product markets. It shows that nonreportable deals may reduce future competition by shutting down overlapping R&D pipelines.

Economic message: The antitrust issue is not only current pricing power. In pharmaceuticals, avoiding review may allow firms to remove future rivals before products reach the market.

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Table VI

Overlapping Pharmaceutical Projects and Nonreportable M&As

This table presents results from OLS regressions of pharmaceutical projects on an indicator for whether the deal was reportable or nonreportable to the antitrust regulators. The main variable of interest in Panel A, *Nonreportable*, is an indicator that assumes the value of one if the target firm's assets are below the size-of-person (SoP) asset threshold, and zero otherwise. In Panel B, the main variable interest in columns (1) to (3) is *Nonreportable*; in columns (4) and (5), it is the interaction term *Nonreportable* × *EconImportance*. In column (4), *EconImportance* assumes the value of one if the target's drug project is in Phase 3 of trials, and zero otherwise, and in column (5), *EconImportance* assumes the value of one if there are three or fewer competitors developing an overlapping drug that matches the target's project, and zero otherwise. The main variable of interest in Panel C, *Nonreportable* × *AcquiredProject*, is an interaction term that assumes the value of one when an overlapping project is acquired in a nonreportable deal, and zero otherwise. In columns (1) and (2) of Panel A, the dependent variable, *Pr(ProjectOverlap)*, is an indicator variable that assumes the value of one if the target firm and the acquiring firm have at least one drug development project that directly overlaps, and zero otherwise. In columns (3) and (4) of Panel A, the dependent variable, *ProjectOverlap*, is a continuous variable that measures the proportion of the target firm's drug development projects that overlap with the acquirer's drug development projects. In all columns of Panels B and C, the dependent variable, *ProjectDisc'd*, is an indicator that assumes the value of one if a drug project is discontinued after the acquisition date. All variables are described in Internet Appendix Section VI. For both Panels A and B, we vary the inclusion of fixed effects as follows. In columns (1) and (3) of Panel A, we exclude filing-year fixed effects, and in columns (2) and (4), we include filing-year fixed effects. In column (1) of Panel B, we exclude fixed effects, in columns (2) we include TC fixed effects, and in column (3), we include TC and filing-year fixed effects, respectively. For Panel C, we vary the fixed effects structure across columns. We also vary the inclusion of our control variables, for example, we include control variables in columns (2), (4), (6), and (8). Control variables included in the estimations in Panel C but not reported are *Size*, *Sales*, *Leverage*, *EBITDA/Assets*, *Cash/Assets*, *CashFlow/Assets*, *R&D*, and *Q*. Robust *t*-statistics are reported in parentheses and calculated using standard errors clustered at the filing-year level. *, **, *** represent significance at the 10%, 5%, and 1% level, respectively.

Panel A. Overlapping Projects

Dependent Variable:	(1) *Pr(ProjectOverlap)*	(2) *Pr(ProjectOverlap)*	(3) *ProjectOverlap*	(4) *ProjectOverlap*
Nonreportable	0.082**	0.073*	0.015**	0.012**
	(2.26)	(2.05)	(2.75)	(2.68)
Observations	169	169	169	169
Adjusted R^2	0.016	0.051	0.045	0.051
Filing-Year F/E	N	Y	N	Y

(*Continued*)

Table VI—*Continued*

Panel B. Drug Project-Level Development and Competition

Dependent Variable:	(1) *ProjectDisc'd*	(2) *ProjectDisc'd*	(3) *ProjectDisc'd*	(4) *ProjectDisc'd*	(5) *ProjectDisc'd*
EconImportance:				*Phase 3 Trials*	*High Market Concentration*
Nonreportable	0.148**	0.332**	0.595*	0.597*	0.306
	(2.92)	(2.39)	(2.29)	(2.25)	(1.07)
EconImportance				0.042	−0.088***
				(0.61)	(-6.10)
Noreportable × *EconImportance*				0.643***	0.686***
				(9.44)	(20.33)
Observations	210	210	210	210	210
Adjusted R^2	0.016	0.044	0.088	0.097	0.121
Therapeutic Class F/E	N	Y	Y	Y	Y
Filing-Year F/E	N	N	Y	Y	Y

(*Continued*)

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Table VI—*Continued*

Panel C. Drug Project-Level Development

Dependent Variable:	(1) *ProjectDisc'd*	(2) *ProjectDisc'd*	(3) *ProjectDisc'd*	(4) *ProjectDisc'd*	(5) *ProjectDisc'd*	(6) *ProjectDisc'd*	(7) *ProjectDisc'd*	(8) *ProjectDisc'd*
Nonreportable × *AcquiredProject*	0.161***	0.331**	0.235***	0.414**	0.216***	0.361**	0.282***	0.366*
	(3.51)	(2.26)	(4.60)	(2.82)	(4.31)	(2.52)	(6.44)	(2.08)
Nonreportable	−0.013	−0.026	0.000	−0.007	−0.043	−0.026	−0.002	−0.018
	(−0.38)	(−0.79)	(0.00)	(−0.23)	(−0.87)	(−1.25)	(−0.05)	(−0.64)
AcquiredProject	0.209***	0.058	0.200***	0.028	0.190***	0.074	0.137***	0.074
	(4.76)	(0.41)	(4.17)	(0.19)	(4.15)	(0.52)	(3.40)	(0.42)
Observations	3,504	2,541	3,504	2,541	3,504	2,541	2,658	2,003
Adjusted R^2	0.038	0.065	0.043	0.073	0.071	0.104	0.265	0.328
Controls	N	Y	N	Y	N	Y	N	Y
Therapeutic Class F/E	N	N	Y	Y	Y	Y	N	N
Filing-Year F/E	N	N	N	N	Y	Y	Y	Y
TC-by-MOA F/E	N	N	N	N	N	N	Y	Y

6 Interpretive synthesis

- The paper links three domains: accounting measurement, antitrust screening, and product-market outcomes.
- The regulatory issue arises because the SoP test treats book assets as if they measure firm size, while in intangible-intensive industries book assets increasingly fail to measure economic size.
- The empirical results form a chain: intangibles are omitted; omitted intangibles make deals nonreportable; nonreportable deals have higher premia and positive stock-market reactions; markups rise; overlapping pharmaceutical projects are discontinued.
- The product-market-overlap tests are central because they distinguish anticompetitive interpretations from generic acquisition benefits.
- The policy counterfactual is nuanced: adding intangibles to the SoP test would make more deals reportable, but firms may also manipulate around new thresholds and enforcement agencies need resources to review additional filings.

7 Short summary

- The Hart-Scott-Rodino premerger notification system uses accounting assets to determine whether many deals must be reported.
- GAAP excludes most internally generated intangibles, so intangible-rich targets can appear too small for review.
- Using purchase price allocations, the paper shows that acquired intangibles have become much larger than acquired tangible assets.
- Nonreportable horizontal deals are concentrated in intangible-intensive sectors such as technology and chemicals.
- Deal premia are higher for nonreportable deals, especially when product markets overlap.
- Acquirer and rival stock prices rise around nonreportable product-overlap deals.
- Markups rise after nonreportable product-overlap acquisitions and persist for several years.
- Pharmaceutical nonreportable deals are more likely to involve overlapping drug projects, and acquired overlapping projects are more likely to be discontinued.
- Adding intangibles to the SoP test would generate additional reports and reviews, but enforcement capacity and avoidance behavior remain important constraints.

8 Conclusion

8.1 Core conclusion

Kepler, McClure, and Stewart show that accounting rules can shape antitrust enforcement. Because the SoP test relies on book assets, and book assets largely omit internally generated intangibles, many economically meaningful and potentially anticompetitive mergers avoid ex ante public review.

8.2 What the paper establishes

The paper establishes a tight institutional channel from accounting measurement to regulatory visibility. It documents the scale of omitted intangible assets, validates the SoP threshold using notification data, and shows that nonreportable deals are associated with higher premia, equity gains, markups, and pharmaceutical project discontinuations.

8.3 What the paper does not fully establish

The paper does not claim that every nonreportable deal is anticompetitive or that adding intangibles to the SoP test would mechanically solve merger enforcement. Enforcement capacity, private litigation frictions, threshold manipulation, and legitimate efficiencies all complicate policy design.

8.4 Incremental policy implication

The paper's policy contribution is to show that merger-screening rules must adapt to the intangible economy. If regulators continue to use accounting assets as bright-line thresholds, antitrust enforcement will be biased away from the very sectors where intangible assets and consumer-facing market power are most important.

8.5 Final takeaway

The rise of intangible capital changes more than firm valuation. It changes which transactions are visible to regulators. In an economy increasingly built on brands, software, customer relationships, and R&D pipelines, accounting-based thresholds can unintentionally create a pathway for anticompetitive consolidation.